

Backtesting Pitfalls Map

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Overfitting and look-ahead bias

PROFESSIONAL DEFINITION

The Backtesting Pitfalls Map catalogues the ways historical strategy tests mislead — overfitting, look-ahead bias, survivorship bias and multiple testing — so apparent edges can be scrutinised.

WHO DEVELOPED IT

Associated with Marcos López de Prado's work on the dangers of backtesting in financial machine learning.

WHY IT WAS DEVELOPED

It was developed because AI makes it trivially easy to find spurious patterns, and most spectacular backtests are false discoveries.

FIGURE 1 · BACKTESTING PITFALLS MAP

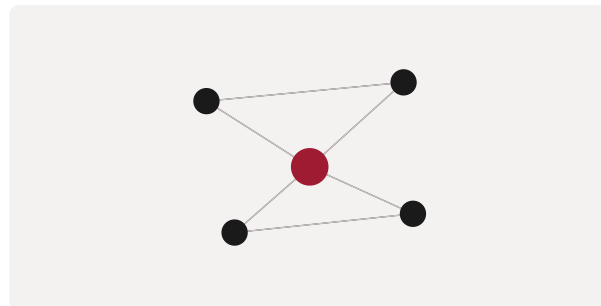


Figure 1. A schematic of the Backtesting Pitfalls Map as applied in BARBM312 (illustrative).

HOW IT IS USED

A backtest is examined against each pitfall; only strategies that survive scrutiny and out-of-sample testing are trusted.

WHEN IT IS USED

When evaluating quantitative and AI-driven investment strategies.

BY WHOM IT IS USED

Quantitative investors, risk and research teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It systematises scepticism toward backtests, a crucial discipline as AI multiplies the risk of false discovery.

RELATED FRAMEWORKS IN THIS COURSE

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Risk-Profiling Matrix · Slibrary 23

Prediction Machines Lens · Slibrary 23

Monte Carlo Cone · Slibrary 23

SOURCES (HARVARD REFERENCING STYLE)

López de Prado, M. (2018) *Advances in Financial Machine Learning*. Hoboken: Wiley.

Bailey, D.H. et al. (2014) 'Pseudo-mathematics and financial charlatanism', *Notices of the AMS*, 61(5).